


Faculty of Information Technology and Artificial Intelligence

# Introduction to System Administration

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# 1. Objectives

The purpose of this practical assignment is to practice the basic Linux commands and become familiar with the Linux Command Line Interface (CLI). The work covers displaying manual pages, managing files and directories, navigating paths, and processing text files from the terminal. These skills are essential for system administration tasks.

# 2. Experimental Environment and Materials

The commands were executed in the terminal (CLI) of Ubuntu Linux running inside a VMware virtual machine. Sample text files and directories, such as fruits.txt, students.txt, longfile.txt and dir1, were created in the Lecture3_Lab working directory and used to test the commands.

# 3. Procedure

The following subsections show the commands that were executed, together with the original terminal screenshots and a short description of each result.

## 3.1 Working Directory Preparation

The mkdir -p command creates a directory (and any missing parent directories), cd changes the current directory, and pwd prints the full path of the current directory. ls -l lists the directory contents in long format.

**Command(s):**

```
mkdir -p ~/Lecture3_Lab
cd ~/Lecture3_Lab
pwd
```

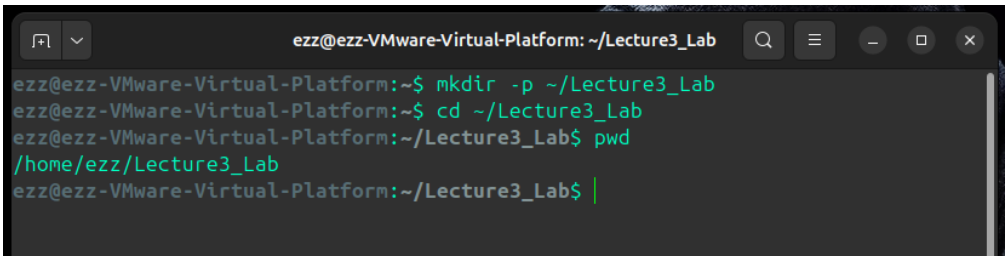



*Figure 1. Creating the Lecture3_Lab directory, entering it, and printing the working directory.*

**Result:** The directory was created and pwd printed /home/ezz/Lecture3_Lab.

**Command(s):**

```
ls -l
```

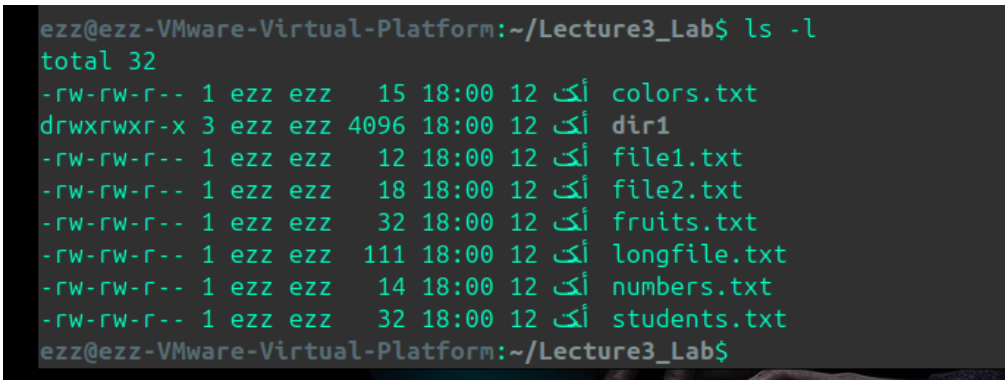



*Figure 2. Long listing of the files prepared in Lecture3_Lab.*

**Result:** The listing shows colors.txt, dir1, file1.txt, file2.txt, fruits.txt, longfile.txt, numbers.txt and students.txt, with dir1 marked as a directory (d).

## 3.2 Manual and Help Commands

The man command opens the manual page of a command. The -k option searches the short descriptions of all manual pages for a keyword, while -K searches the full text of the manual pages.

**Command(s):**

```
man ls
```




*Figure 3. Running man ls.*

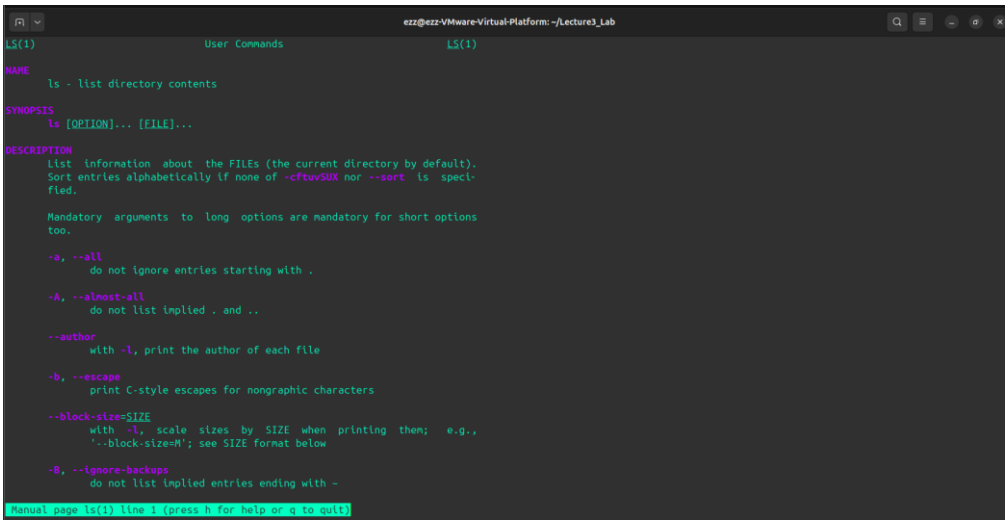



*Figure 4. The ls(1) manual page.*

**Result:** The manual page for ls opened, showing the NAME, SYNOPSIS and DESCRIPTION sections and options such as -a (--all) and -A (--almost-all).

**Command(s):**

```
man 1 intro
```

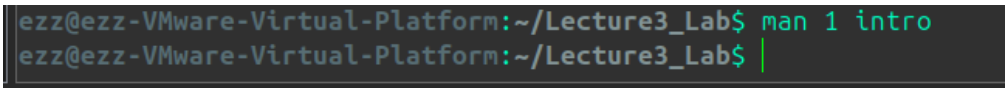



*Figure 5. Opening section 1 of the intro manual page.*




*Figure 6. The intro(1) manual page.*

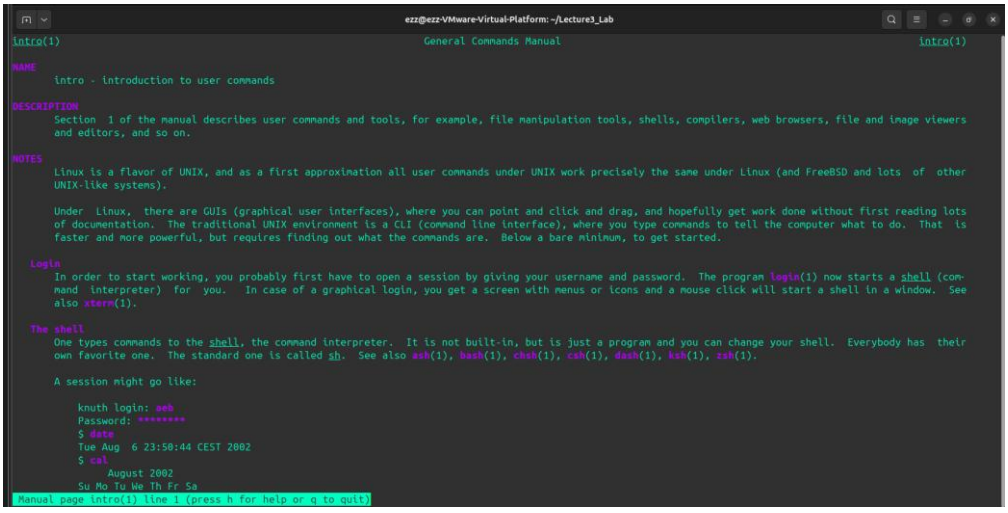



*Figure 7. A second capture of the first screen of the intro(1) manual page.*

**Result:** The intro(1) page opened with the title "intro - introduction to user commands" and described Linux, the CLI and the shell.

**Command(s):**

```
man -k copy
```

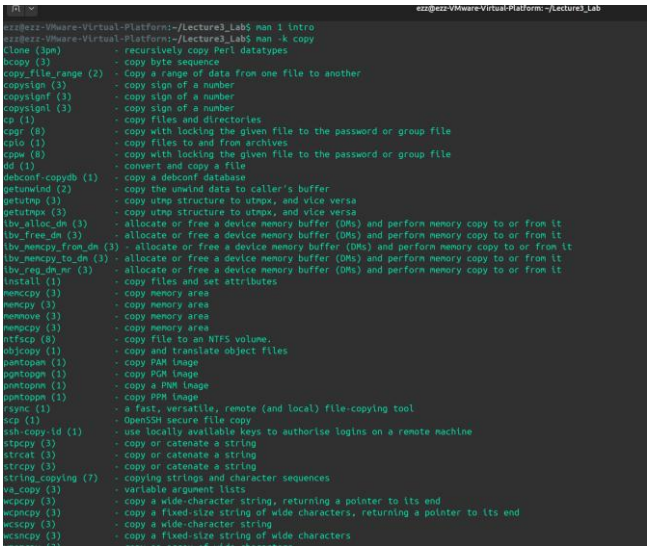



*Figure 8. Searching manual page descriptions for the keyword "copy".*

**Result:** A list of manual pages whose descriptions contain "copy" was displayed, including cp(1), cpio(1), scp(1) and rsync(1).

**Command(s):**

```
man -K password
```

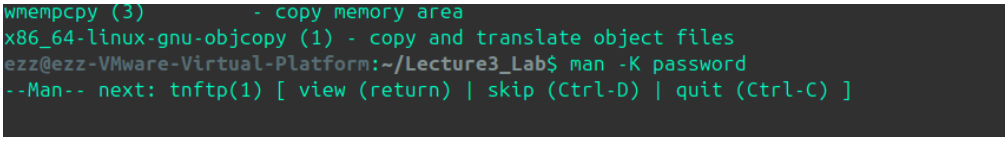
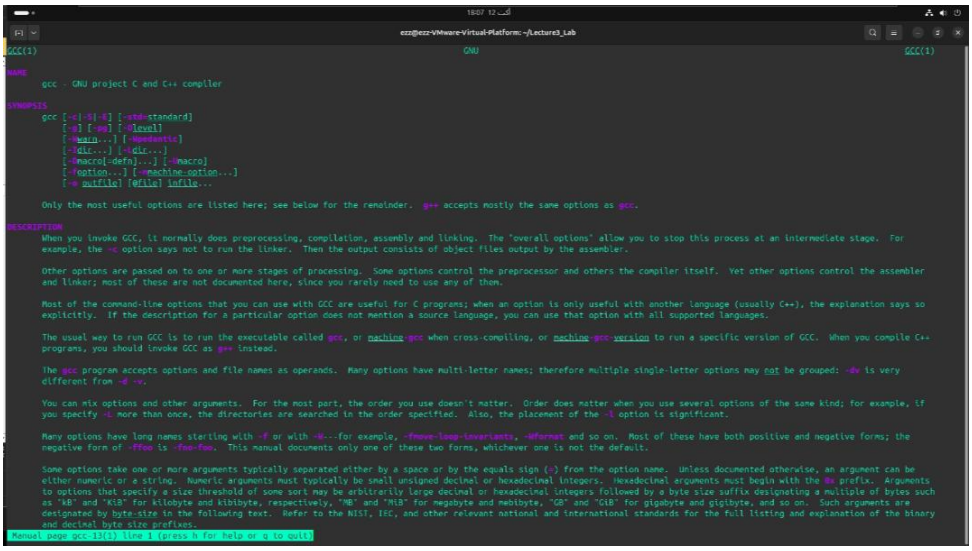



*Figure 9. End of the man -k copy output, followed by man -K password.*

**Result:** The man -K password search started and stopped at tnftp(1), showing the prompt "view (return) | skip (Ctrl-D) | quit (Ctrl-C)" and waiting for input.

**Command(s):**

```
man 1 intro
```



## 3.3 Account-Related Commands

The passwd command is used to change the password of the current user account.

**Command(s):**

```
passwd
```

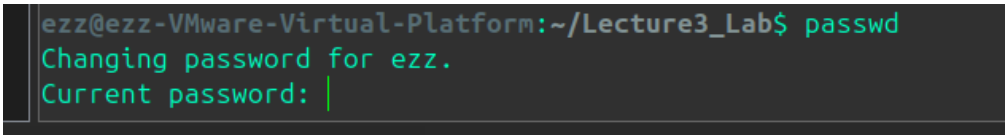



*Figure 11. Starting the passwd command.*

**Result:** The terminal displayed "Changing password for ezz." and the prompt "Current password:". No further steps are visible in the screenshot.

## 3.4 Copying Files and Directories

The cp command copies files and directories. The -r option copies a directory together with all its contents.

**Command(s):**

```
cp file1.txt copy1.txt
ls -l
```

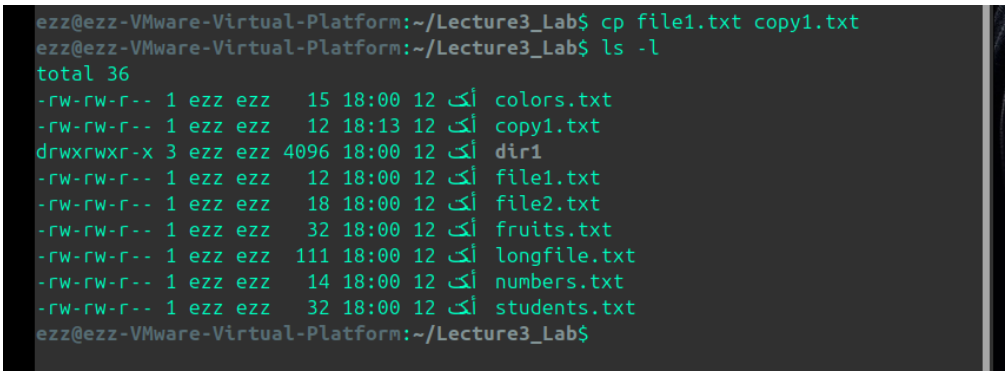



*Figure 12. Copying file1.txt to copy1.txt and verifying it with ls -l.*

**Result:** copy1.txt appeared in the listing with the same size (12 bytes) as file1.txt.

**Command(s):**

```
cp file1.txt dir1/
ls dir1
```

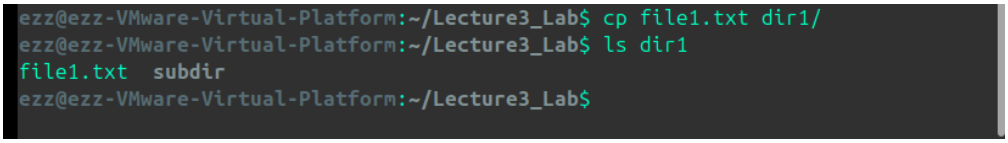



*Figure 13. Copying file1.txt into the dir1 directory.*

**Result:** ls dir1 shows file1.txt and subdir, so the file was copied into dir1.

**Command(s):**

```
cp -r dir1 dir2
ls -R dir2
```

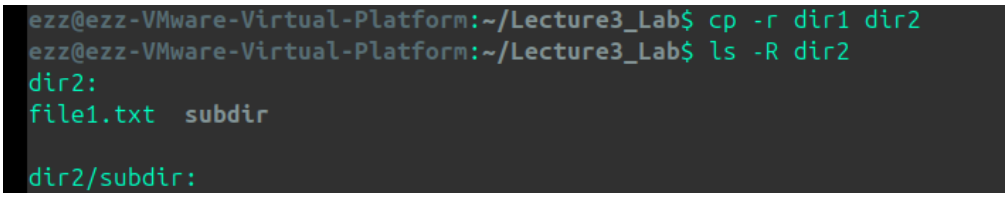



*Figure 14. Copying dir1 recursively to dir2 and listing it with ls -R.*

**Result:** dir2 contains file1.txt and subdir. The ls -R output is cut off after the "dir2/subdir:" heading in the screenshot.

## 3.5 Moving and Renaming Files

The mv command moves a file to another location or renames it when the source and destination are in the same directory.

**Command(s):**

```
mv copy1.txt renamed.txt
ls
```

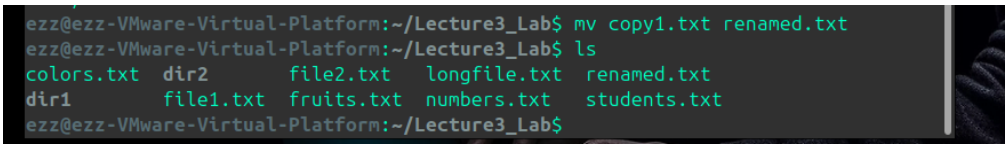



*Figure 15. Renaming copy1.txt to renamed.txt.*

**Result:** renamed.txt now appears in the listing and copy1.txt is no longer listed.

## 3.6 Viewing File Contents

The cat command prints a whole file, while more and less display a long file one screen at a time.

**Command(s):**

```
cat file1.txt
```

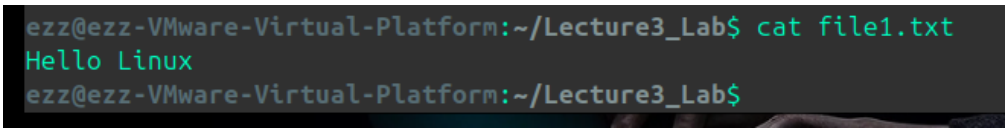



*Figure 16. Displaying file1.txt with cat.*

**Result:** The file contains one line: "Hello Linux".

**Command(s):**

```
more longfile.txt
```

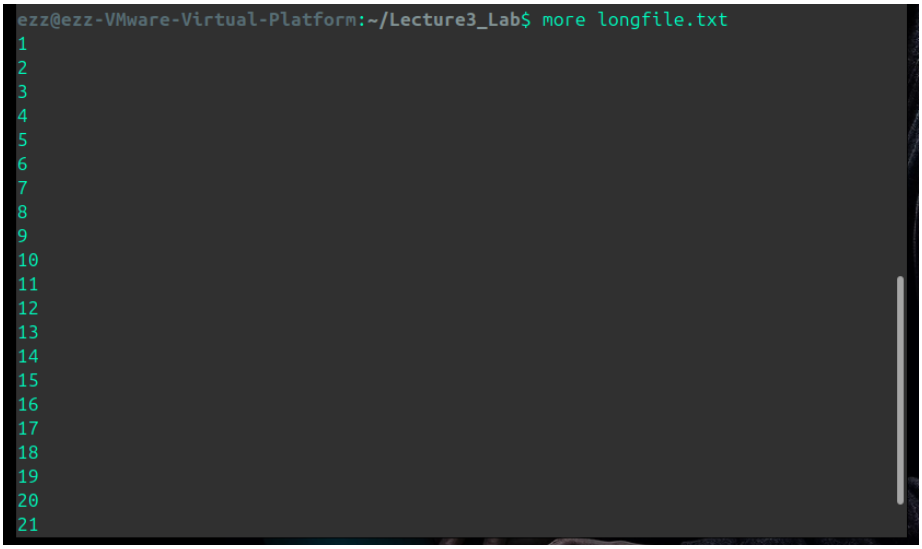



*Figure 17. Viewing longfile.txt page by page with more.*

**Result:** The numbers 1 to 21 are visible on the first screen of the file.

**Command(s):**

```
less longfile.txt
```

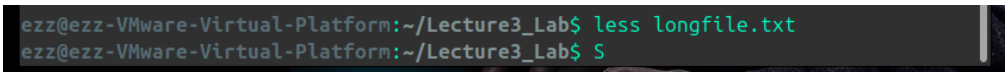



*Figure 18. Running less on longfile.txt.*

**Result:** The screenshot shows the command and the prompt after it; the paged content itself was not captured.

## 3.7 Directory Management

The mkdir command creates a directory. The cd command moves between directories, where ".." means the parent directory and "~" means the home directory.

**Command(s):**

```
ls
```

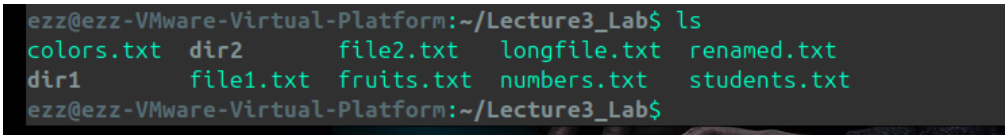



*Figure 19. Listing the directory before creating testdir.*

**Result:** The listing shows colors.txt, dir1, dir2, file1.txt, file2.txt, fruits.txt, longfile.txt, numbers.txt, renamed.txt and students.txt.

**Command(s):**

```
mkdir testdir
ls
```

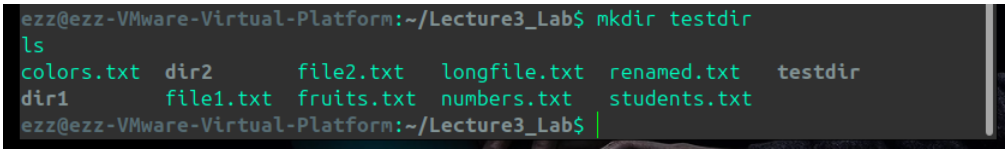



*Figure 20. Creating the testdir directory and listing the result.*

**Result:** testdir now appears in the listing.

**Command(s):**

```
cd testdir
pwd
```

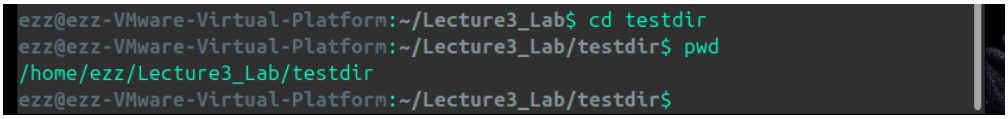



*Figure 21. Entering testdir.*

**Result:** The prompt changed and pwd printed /home/ezz/Lecture3_Lab/testdir.

**Command(s):**

```
cd ..
pwd
```

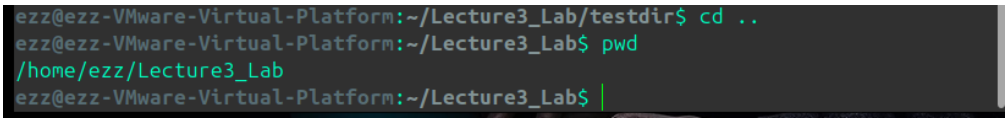



*Figure 22. Returning to the parent directory.*

**Result:** pwd printed /home/ezz/Lecture3_Lab.

**Command(s):**

```
cd ~
pwd
```

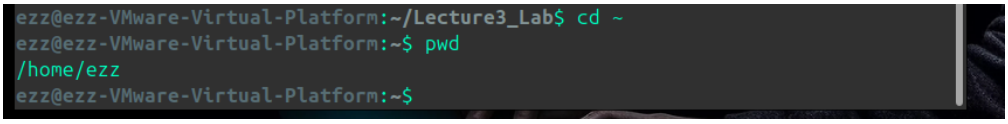



*Figure 23. Going to the home directory.*

**Result:** pwd printed /home/ezz.

## 3.8 Removing Files and Directories

The rm command deletes files, rmdir removes empty directories, and rm -r removes a directory with its contents. The -d option of ls lists a directory itself instead of its contents.

**Command(s):**

```
echo "Delete me" > delete.txt
ls delete.txt
rm delete.txt
ls delete.txt
```

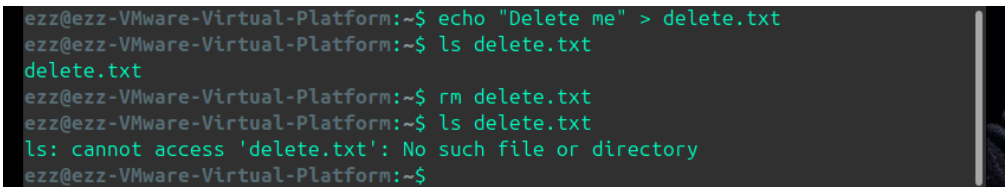



*Figure 24. Creating, deleting and checking delete.txt.*

**Result:** The file was listed before rm; afterwards ls reported "No such file or directory", confirming it was deleted.

**Command(s):**

```
mkdir emptydir
rmdir emptydir
ls -d emptydir
```

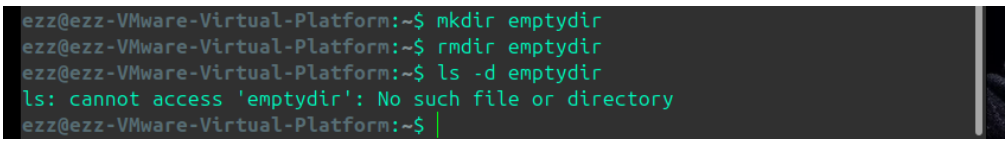



*Figure 25. Creating and removing an empty directory.*

**Result:** After rmdir, ls -d reported that emptydir does not exist.

**Command(s):**

```
mkdir -p deletefolder/inside
touch deletefolder/inside/test.txt
ls -R deletefolder
rm -r deletefolder
ls -d deletefolder
```

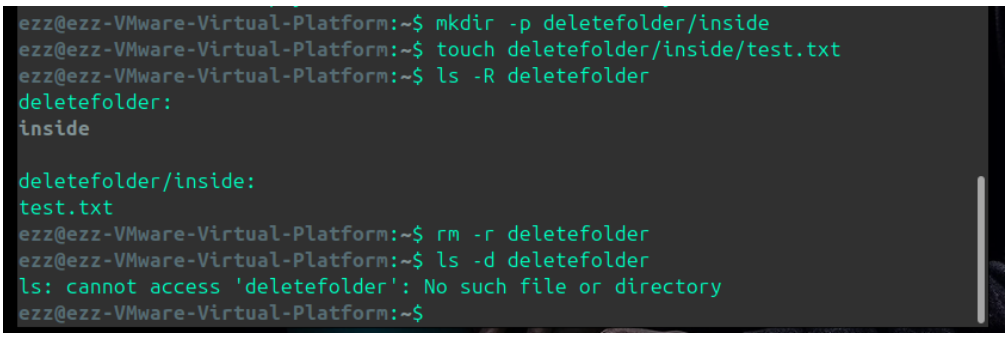



*Figure 26. Creating a nested folder with a file, then removing it with rm -r.*

**Result:** ls -R showed test.txt inside deletefolder/inside; after rm -r, deletefolder no longer existed.

## 3.9 Relative and Absolute Paths

In a relative path, "." means the current directory and ".." means the parent directory. An absolute path starts from the root directory (/). The command pwd -P prints the physical absolute path of the current directory.

**Command(s):**

```
cd ~/Lecture3_Lab
pwd
ls.
ls ..
```

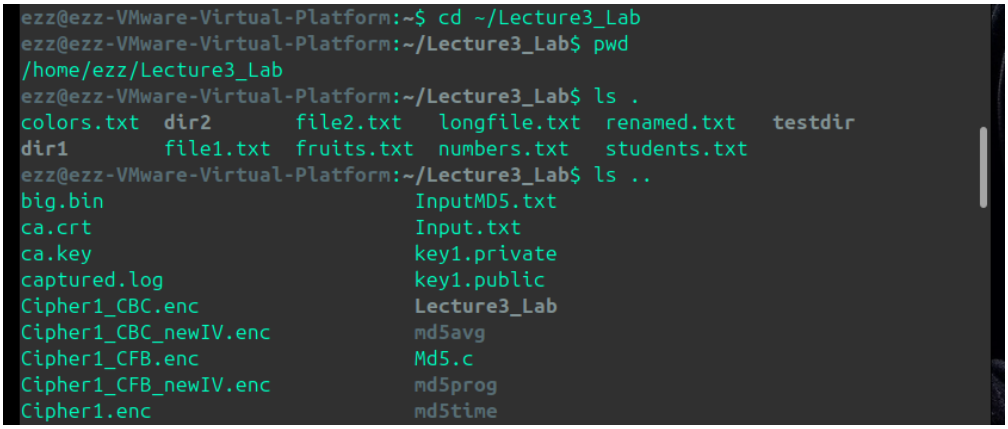



*Figure 27. Listing the current directory (.) and its parent directory (..).*

**Result:** ls . listed the Lecture3_Lab files and directories, while ls .. listed the contents of the home directory (only part of that list is visible).

**Command(s):**

```
ls ~
```

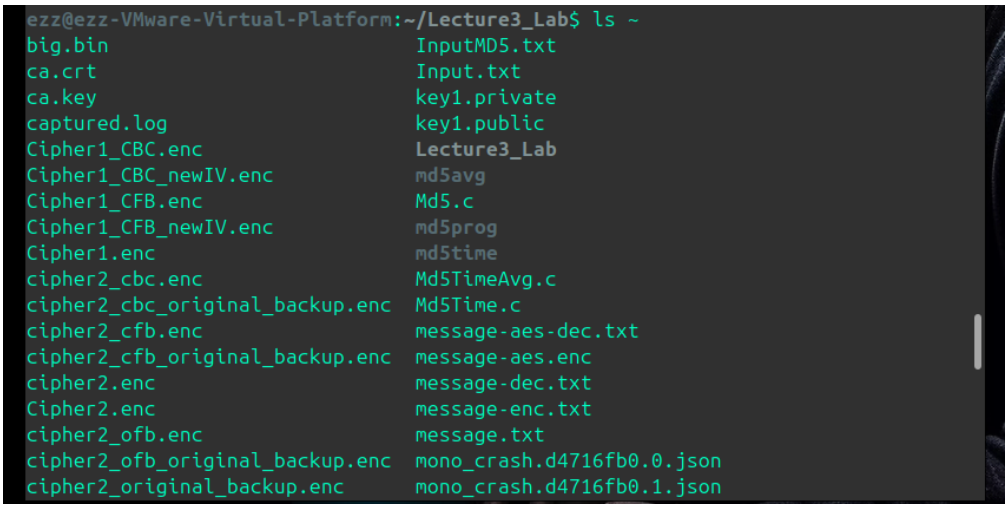



*Figure 28. Listing the home directory using ~.*

**Result:** The output lists the contents of the home directory, including Lecture3_Lab, and is the same location that ls .. showed.

**Command(s):**

```
cd "$(pwd -P)/dir1"
pwd
```

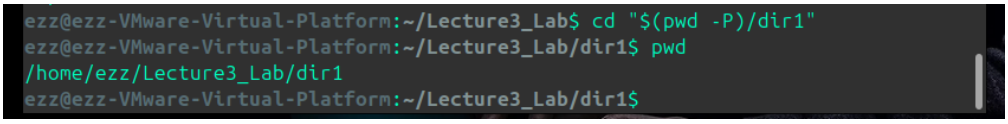



*Figure 29. Changing to dir1 using an absolute path built with pwd -P.*

**Result:** pwd printed /home/ezz/Lecture3_Lab/dir1.

## 3.10 Head, Tail, and Word Count

The head and tail commands show the first and last lines of a file (the number gives how many lines). The wc command counts lines (-l), words (-w) and bytes (-c).

**Command(s):**

```
head -5 longfile.txt
```

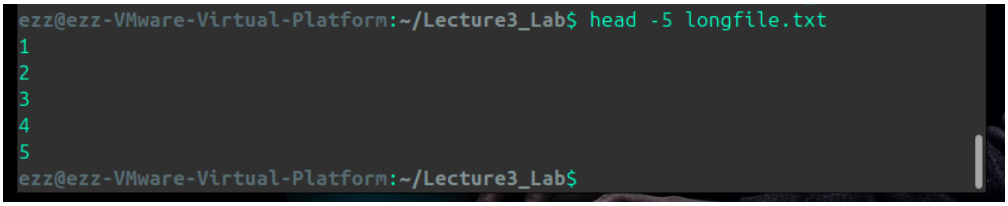



*Figure 30. Displaying the first 5 lines of longfile.txt.*

**Result:** Lines 1 to 5 were printed.

**Command(s):**

```
head -10 longfile.txt
```

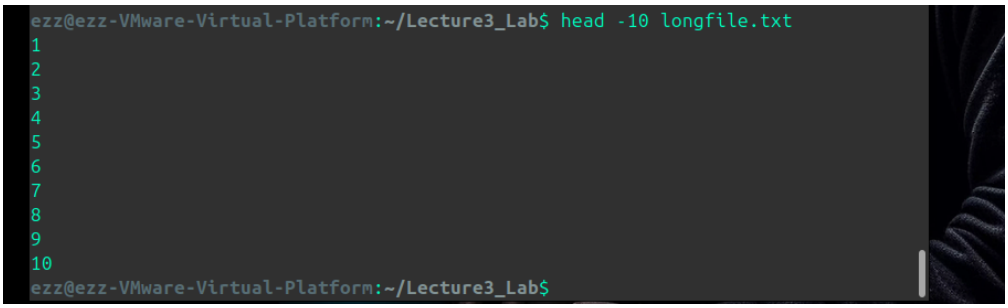



*Figure 31. Displaying the first 10 lines of longfile.txt.*

**Result:** Lines 1 to 10 were printed.

**Command(s):**

```
tail -5 longfile.txt
tail -10 longfile.txt
```

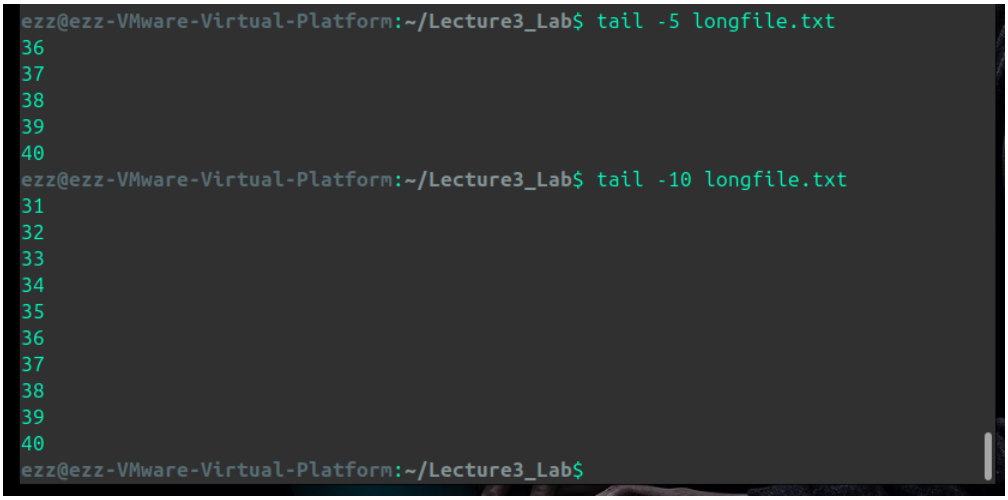



*Figure 32. Displaying the last 5 and last 10 lines of longfile.txt.*

**Result:** The last 5 lines were 36 to 40 and the last 10 lines were 31 to 40.

**Command(s):**

```
wc longfile.txt
wc -l longfile.txt
wc -w longfile.txt
wc -c longfile.txt
```

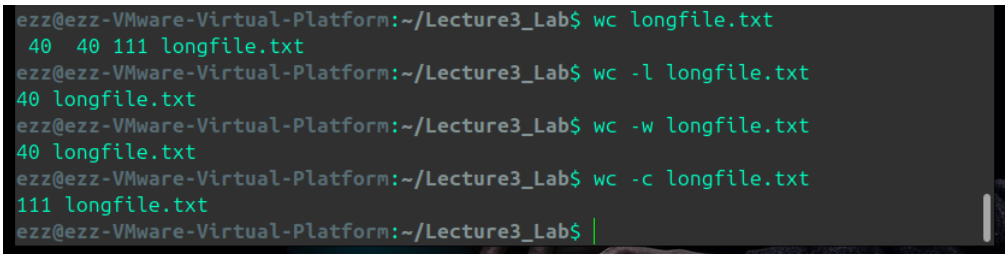



*Figure 33. Counting lines, words and bytes in longfile.txt.*

**Result:** The file has 40 lines, 40 words and 111 bytes.

## 3.11 Sorting and Removing Duplicates

The sort command orders lines alphabetically; -r reverses the order, -t sets the field separator, -k selects the sort field and -n sorts numerically. The uniq command removes repeated adjacent lines, and -c counts the repetitions.

**Command(s):**

```
sort fruits.txt
sort -r fruits.txt
sort -t: -n -k3 students.txt
```

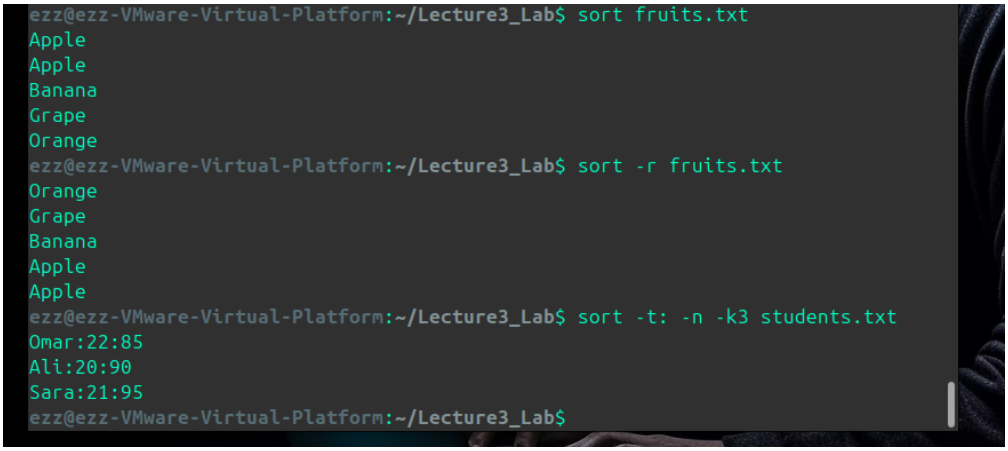



*Figure 34. Sorting fruits.txt normally and in reverse, and sorting students.txt by the third field.*

**Result:** fruits.txt was sorted from Apple to Orange and then from Orange to Apple. students.txt was sorted by the third field in ascending order (85, 90, 95).

**Command(s):**

```
uniq fruits.txt
sort fruits.txt | uniq
sort fruits.txt | uniq -c
```

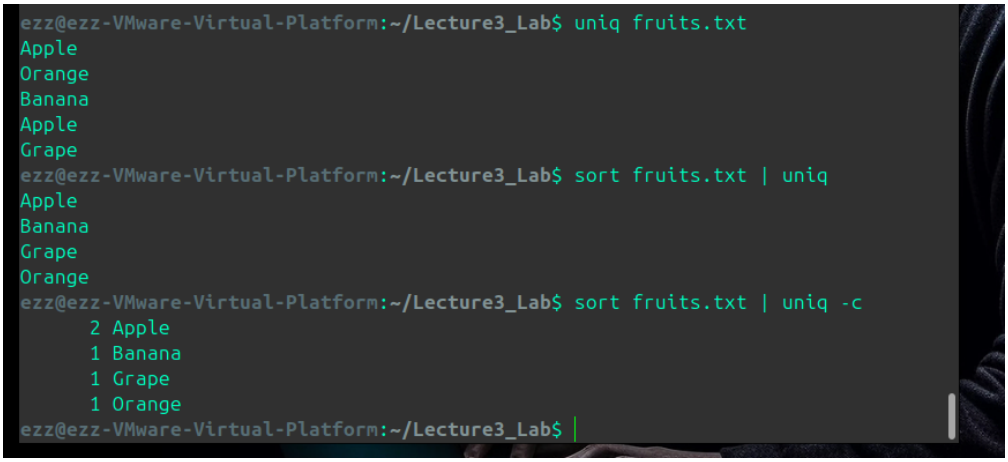



*Figure 35. Removing and counting duplicate lines with uniq.*

**Result:** uniq alone kept both Apple lines because they were not adjacent; after sorting, duplicates were removed, and uniq -c showed that Apple appears 2 times and Banana, Grape and Orange once each.

## 3.12 Extracting Fields Using cut

The cut command extracts selected parts of each line. The -d option sets the delimiter and -f selects the field number.

**Command(s):**

```
cut -d: -f1 students.txt
cut -d: -f2 students.txt
cut -d: -f3 students.txt
```

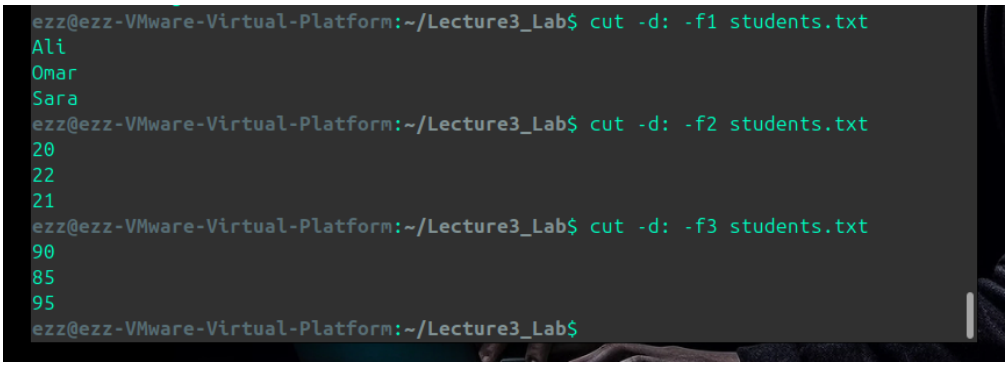



*Figure 36. Extracting the first, second and third fields of students.txt.*

**Result:** Field 1 gave the names (Ali, Omar, Sara), field 2 gave 20, 22 and 21, and field 3 gave 90, 85 and 95.

## 3.13 Searching Text Using grep

The grep command prints the lines that contain a given pattern. The -i option ignores letter case and -n shows the line numbers of the matches.

**Command(s):**

```
grep "Ali" students.txt
```

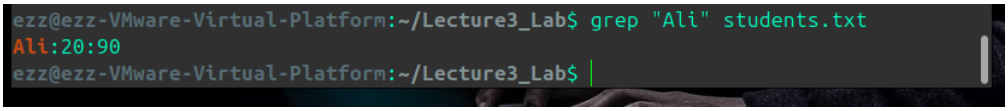



*Figure 37. Searching for "Ali" in students.txt.*

**Result:** The matching line Ali:20:90 was printed.

**Command(s):**

```
grep "Ali" students.txt
grep "Apple" fruits.txt
grep -i "apple" fruits.txt
grep -n "Apple" fruits.txt
```

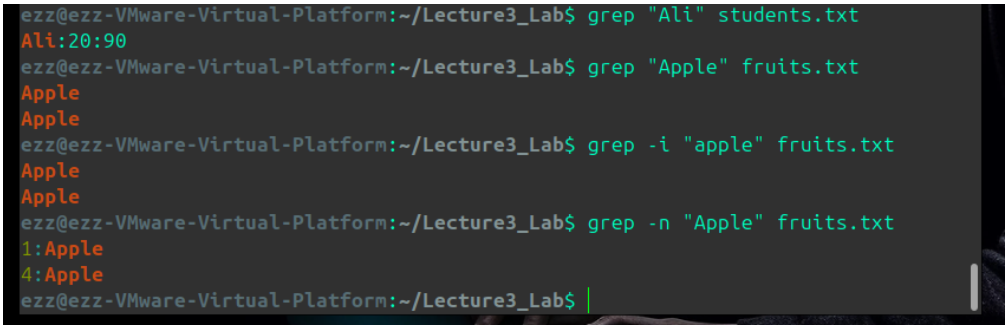



*Figure 38. Searching students.txt and fruits.txt with grep, -i and -n.*

**Result:** Both Apple lines were found with and without -i, and -n showed them at lines 1 and 4.

## 4. Conclusions

In this assignment, basic Linux commands were practiced in the Terminal on Ubuntu. Files and directories were created, copied, moved, renamed and removed, and the pwd and cd commands were used to navigate between directories with relative and absolute paths. Text files were viewed and processed using head, tail, wc, sort, uniq, cut and grep. The assignment showed how powerful and efficient the command line is for everyday file management and text filtering.

## 5. References

[1] Introduction to System Administration, Lecture 3 – Linux Commands, Faculty of Information Technology and Artificial Intelligence (course material).